

MATH 4720/MSSC 5720 Spring 2026 Quiz 1

Feb 5, 2026

Name: _____ Student ID: _____

1. **Mean and standard deviation (SD):** For cases (1) and (2), compare Data A and Data B based on their mean and SDs. You **don't** need to calculate these statistics, but compare A and B by stating
 - (i) which one has a larger mean or they have the same mean.
 - (ii) which one has a larger SD or they have the same SD.
- (1) (A) -20, 0, 15, 25
(B) -40, 0, 15, 25
- (2) (A) 100, 200, 300, 400, 500
(B) 0, 50, 300, 550, 600

Answer:

2. **Skewness:** Instagram data indicate that 50% of Instagram users have 80 or more followers, and that the average follower count of users is 290. What do these findings suggest about the shape (right-skewed, left-skewed, symmetric) of the distribution of number of followers of Instagram users? Briefly explain.

Answer:

3. **Probability:** Which probability rule below means that event A and event B are independent? Choose one, and no need to provide reasoning.
 - (1) $P(A \cup B) = P(A) + P(B)$
 - (2) $P(A \cap B) = P(A) \times P(B)$
 - (3) $P(A | B) = P(B)$

Answer: